20/2/25 Exp-8 NMAP to Discover Live Host using Nmap Scans on the TryHackme

Aim:-

To study the working of the Nmap tool and discover live hosts in a network using different host discovery techniques such as ARP Scanning, ICMP Scanning, TCP Scanning, and UDP Scanning on the TryHackme Platform. The experiment also aims to analyze the behaviour of different protocols and compare the result obtained from various scanning methods.

Steps:-

1. Open the TryHackme Platform and Select the Nmap room.
2. Select the Network topology and identify the Sources and Destination hosts.
3. Use ARP requests to discover the MAC address of devices available in the local network.
4. Observe how ARP Packets are broadcast within the network and identify the devices that receive the request.
5. Use the ICMP Protocol to send Ping requests b/w two System and observe the responses.
6. Analyze the Packet flow and identify the exchange of ARP Packets before ICMP communication.
7. Perform ARP host discovery using the Nmap command and identify all active hosts in the local network.

8. Perform ICMP timestamp ping scanning and observe the hosts that responded to the scan.
9. Perform ICMP timestamp scanning and analyze the response generated by the target system.
10. Perform ICMP address mask scanning and analyze the results with Wireshark.
11. Perform TCP SYN scanning to identify live hosts by analyzing SYN and ACK packets.
12. Perform TCP ACK scanning to verify the accessibility of the target hosts.
13. Perform UDP scanning and observe how Nmap identifies active hosts through UDP packets.
14. Perform a reverse DNS lookup to obtain the hostnames of the discovered systems.
15. Compare the results obtained from ARP, ICMP, TCP and UDP scanning techniques.
16. Analyze the advantage and limitation of each scanning method.

Result:-

Thus, the hosts in the network were successfully identified using different Nmap scanning techniques. ARP scanning was found to be effective for local devices while ICMP, TCP and UDP scanning techniques were useful for identifying active hosts. The experiment also demonstrated the effectiveness of Nmap tool in network discovery and analysis.